PHYDSC402T — Classical Mechanics

Unit 4: Hamilton-Jacobi Theory and Action-Angle Variables

7th Semester, Assam University (FYUG) — Haflong Government College

Paper: PHYDSC402T, Full Marks 100 [ESE 70 + CCA 30] | Unit 4: 13 lectures

Notes 10 of 20 — drafted by gemini-3.6-flash (Kaggle Model Proxy) from the unit syllabus only, without external sources, and checked against the syllabus.

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Unit syllabus (as in the syllabus file)

1. Hamilton-Jacobi theory, Hamilton's characteristic function, Harmonic oscillator in Hamilton-Jacobi method, separation of variables in Hamilton-Jacobi equation.
2. Action and angle variables, Kepler problem in action-angle variables.

Top 2 high-mark topics in this unit

1. Hamilton-Jacobi equation, Hamilton's characteristic function and separation of variables in the Hamilton-Jacobi equation — 13 lectures, the heaviest unit in the paper, and a full derivation topic.
2. Harmonic oscillator by the Hamilton-Jacobi method and by action-angle variables, including frequency from action variable J — the only fully covered HJ/action-angle topic in the sources, repeated in Christ College and St. Aloysius papers.

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2. Hamilton's Characteristic Function
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4. Harmonic Oscillator in the Hamilton-Jacobi Method
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6. Kepler Problem in Action-Angle Variables
7. Questions — Short form, then Long form (no AU questions found for this unit)
8. Sources

1. Hamilton-Jacobi Theory

Hamilton-Jacobi theory provides a formulation of classical mechanics in which the equations of motion are transformed into a single first-order partial differential equation. Consider a mechanical system described by n generalized coordinates $q = (q_1, q_2, \ldots, q_n)$, their corresponding generalized momenta $p = (p_1, p_2, \ldots, p_n)$, and explicit time t, with Hamiltonian $H(q, p, t)$.

The central objective is to find a canonical transformation from the original phase space variables (q_i, p_i) to a new set of constant canonical variables (Q_i, P_i). For the new variables to be constants of motion, their time derivatives must vanish identically:

$$\dot{Q}_i = 0$$

$$\dot{P}_i = 0$$

Hamilton's equations of motion for the new canonical variables (Q_i, P_i) in terms of the transformed Hamiltonian $K(Q, P, t)$ are expressed as:

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}$$

$$\dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

To guarantee that $\dot{Q}_i = 0$ and $\dot{P}_i = 0$ for all i, it is sufficient to demand that the transformed Hamiltonian vanishes identically:

$$K = 0$$

A canonical transformation can be generated using a generating function of the second type, denoted as $F_2(q, P, t)$. In Hamilton-Jacobi theory, this generating function is denoted by $S(q, \alpha, t)$ and is called Hamilton's principal function, where the new constant momenta P_i are set equal to constants of integration α_i:

$$P_i = \alpha_i$$

The transformation equations relating the old variables (q_i, p_i) and the new variables $(Q_i, P_i = \alpha_i)$ via Hamilton's principal function $S(q, \alpha, t)$ are given by:

$$p_i = \frac{\partial S}{\partial q_i}$$

$$Q_i = \frac{\partial S}{\partial \alpha_i}$$

Since the new coordinates Q_i are also constants of motion, they are set equal to constant parameters β_i:

$$Q_i = \beta_i$$

$$\beta_i = \frac{\partial S}{\partial \alpha_i}$$

The general relation linking the original Hamiltonian $H(q, p, t)$, the transformed Hamiltonian $K(Q, P, t)$, and the generating function $S(q, \alpha, t)$ is:

$$K = H + \frac{\partial S}{\partial t}$$

Applying the requirement that the transformed Hamiltonian must vanish, $K = 0$, yields:

$$H + \frac{\partial S}{\partial t} = 0$$

Substituting the expressions for the generalized momenta $p_i = \frac{\partial S}{\partial q_i}$ into the original Hamiltonian gives the Hamilton-Jacobi equation:

$$H\left(q_1, \ldots, q_n, \frac{\partial S}{\partial q_1}, \ldots, \frac{\partial S}{\partial q_n}, t\right) + \frac{\partial S}{\partial t} = 0$$

This is a first-order, non-linear partial differential equation in $n + 1$ variables $(q_1, \ldots, q_n, t)$ for the unknown function S. A complete solution to this partial differential equation contains n independent non-additive constants of integration $\alpha_1, \ldots, \alpha_n$, along with an unessential additive constant. Once Hamilton's principal function $S(q, \alpha, t)$ is obtained, the algebraic equations $\beta_i = \frac{\partial S}{\partial \alpha_i}$ can be inverted to yield the position coordinates q_i as functions of time t and the $2n$ constants α_i and β_i.

2. Hamilton's Characteristic Function

When the Hamiltonian $H(q, p)$ does not explicitly depend on time t, the explicit time derivative of H vanishes:

$$\frac{\partial H}{\partial t} = 0$$

Under this condition, the total energy of the conservative system is a constant of motion equal to a constant E:

$$H(q, p) = E$$

Because the Hamiltonian is conservative, the explicit time dependence of Hamilton's principal function $S(q, \alpha, t)$ can be separated. The function $S(q, \alpha, t)$ is expressed as the difference between a time-independent function $W(q, \alpha)$ and a term linear in time:

$$S(q, \alpha, t) = W(q, \alpha) - Et$$

Here, $W(q, \alpha)$ is known as Hamilton's characteristic function, where $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ represents the set of integration constants.

To derive the differential equation satisfied by $W(q, \alpha)$, the partial time derivative of $S(q, \alpha, t)$ is evaluated:

$$\frac{\partial S}{\partial t} = \frac{\partial}{\partial t}\left[W(q,\alpha) - Et\right]$$

$$\frac{\partial S}{\partial t} = -E$$

The generalized momenta p_i are calculated from the gradient of $S(q,\alpha,t)$ with respect to the coordinates q_i:

$$p_i = \frac{\partial S}{\partial q_i} = \frac{\partial}{\partial q_i}\left[W(q,\alpha) - Et\right]$$

$$p_i = \frac{\partial W}{\partial q_i}$$

Substituting the expression for p_i and the partial time derivative $\frac{\partial S}{\partial t} = -E$ into the general Hamilton-Jacobi equation $H\left(q, \frac{\partial S}{\partial q}\right) + \frac{\partial S}{\partial t} = 0$ gives:

$$H\left(q_1, \ldots, q_n, \frac{\partial W}{\partial q_1}, \ldots, \frac{\partial W}{\partial q_n}\right) - E = 0$$

Rearranging this equation yields the time-independent Hamilton-Jacobi equation for Hamilton's characteristic function:

$$H\left(q_1, \ldots, q_n, \frac{\partial W}{\partial q_1}, \ldots, \frac{\partial W}{\partial q_n}\right) = E$$

In this formulation, one of the constants of integration α_1 is chosen to be the total energy E:

$$\alpha_1 = E$$

The remaining $n-1$ constants of integration are $\alpha_2, \ldots, \alpha_n$. The new constant coordinate variables β_i are obtained by differentiating $S(q,\alpha,t)$ with respect to α_i:

$$\beta_i = \frac{\partial S}{\partial \alpha_i}$$

For the first constant $\alpha_1 = E$, the corresponding canonical coordinate β_1 is:

$$\beta_1 = \frac{\partial S}{\partial E} = \frac{\partial}{\partial E}\left[W(q,\alpha) - Et\right]$$

$$\beta_1 = \frac{\partial W}{\partial E} - t$$

Rearranging this relation gives the time t explicitly as a function of the coordinates q_i:

$$t + \beta_1 = \frac{\partial W}{\partial E}$$

For the remaining constants α_i (where $i = 2, 3, \ldots, n$), the transformation equations are given by:

$$\beta_i = \frac{\partial S}{\partial \alpha_i} = \frac{\partial}{\partial \alpha_i}\left[W(q,\alpha) - Et\right]$$

$$\beta_i = \frac{\partial W}{\partial \alpha_i}$$

These $n-1$ equations determine the geometric path of the system trajectory in space, independent of time, while the equation $t + \beta_1 = \frac{\partial W}{\partial E}$ specifies the motion along that trajectory as a function of time.

3. Separation of Variables in the Hamilton-Jacobi Equation

When a conservative mechanical system with n degrees of freedom has a Hamiltonian $H(q_1, \ldots, q_n, p_1, \ldots, p_n)$ that does not depend explicitly on time t, the total energy of the system is conserved and equals a constant E. In this case, Hamilton's principal function $S(q_1, \ldots, q_n, \alpha_1, \ldots, \alpha_n, t)$ can be separated into a time-independent part and a time-dependent part according to the relation

$$S(q_1, \ldots, q_n, \alpha_1, \ldots, \alpha_n, t) = W(q_1, \ldots, q_n, \alpha_1, \ldots, \alpha_n) - Et$$

where $W(q_1, \ldots, q_n, \alpha_1, \ldots, \alpha_n)$ is Hamilton's characteristic function, and $\alpha_1, \alpha_2, \ldots, \alpha_n$ are n independent integration constants, with one constant typically chosen as the total energy $E = \alpha_1$.

Substituting the partial derivatives of S with respect to coordinates into the time-dependent Hamilton-Jacobi equation yields the time-independent Hamilton-Jacobi equation:

$$H\left(q_1, \ldots, q_n, \frac{\partial W}{\partial q_1}, \ldots, \frac{\partial W}{\partial q_n}\right) = E$$

Additive Separation of Hamilton's Characteristic Function

The method of additive separation of variables applies when Hamilton's characteristic function W can be expressed as a sum of n independent functions, each depending on only a single generalized coordinate q_i:

$$W(q_1, q_2, \ldots, q_n, \alpha_1, \ldots, \alpha_n) = \sum_{i=1}^{n} W_i(q_i, \alpha_1, \ldots, \alpha_n)$$

Under this additive form, the generalized momentum corresponding to the k-th coordinate is given by the partial derivative of W with respect to q_k:

$$p_k = \frac{\partial W}{\partial q_k} = \frac{dW_k(q_k)}{dq_k}$$

The time-independent Hamilton-Jacobi equation then takes the form:

$$H\left(q_1, \ldots, q_n, \frac{dW_1}{dq_1}, \ldots, \frac{dW_n}{dq_n}\right) = E$$

Conditions for Separability

A Hamiltonian H is said to be separable in a given set of generalized coordinates if it can be written such that one coordinate, say q_1, and its corresponding partial derivative $\frac{dW_1}{dq_1}$ appear in a isolated sub-function Φ_1. The equation can then be rearranged as:

$$\Phi_1\left(q_1, \frac{dW_1}{dq_1}\right) = \Phi_2\left(q_2, \ldots, q_n, \frac{dW_2}{dq_2}, \ldots, \frac{dW_n}{dq_n}, E\right)$$

The left-hand side of this equality depends solely on q_1, while the right-hand side depends on the remaining coordinates $q_2, \ldots, q_n$. Because q_1 can be varied independently of $q_2, \ldots, q_n$, both sides must be identically equal to a constant, denoted by α_1:

$$\Phi_1\left(q_1, \frac{dW_1}{dq_1}\right) = \alpha_1$$

This yields an ordinary differential equation for $W_1(q_1)$:

$$\frac{dW_1}{dq_1} = f_1(q_1, \alpha_1)$$

Integrating both sides with respect to q_1 gives the function W_1:

$$W_1(q_1, \alpha_1) = \int f_1(q_1, \alpha_1)\, dq_1$$

The remaining equation for the rest of the coordinates becomes:

$$\Phi_2\left(q_2, \ldots, q_n, \frac{dW_2}{dq_2}, \ldots, \frac{dW_n}{dq_n}, E\right) = \alpha_1$$

Repeating this process sequentially for $q_2, q_3, \ldots, q_n$ isolates each coordinate into an independent ordinary differential equation:

$$\frac{dW_k}{dq_k} = f_k(q_k, \alpha_1, \alpha_2, \ldots, \alpha_k)$$

Integrating each of these equations yields the separated functions:

$$W_k(q_k, \alpha_1, \ldots, \alpha_n) = \int f_k(q_k, \alpha_1, \ldots, \alpha_n)\, dq_k$$

Summing all individual functions W_k yields the complete Hamilton characteristic function:

$$W(q_1, \ldots, q_n, \alpha_1, \ldots, \alpha_n) = \sum_{k=1}^{n} \int f_k(q_k, \alpha_1, \ldots, \alpha_n)\, dq_k$$

Thus, the condition for separability requires that the Hamiltonian structure allows complete isolation of each coordinate-momentum pair (q_k, p_k) into individual ordinary differential equations containing n independent

separation constants $\alpha_1, \alpha_2, \ldots, \alpha_n$.

4. Harmonic Oscillator in the Hamilton-Jacobi Method

Consider a one-dimensional simple harmonic oscillator of mass m and angular frequency ω. The position of the particle is given by the coordinate q, and its momentum is p.

The Hamiltonian of the system is the sum of its kinetic and potential energies:

$$H(q,p) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$

Because the Hamiltonian H does not explicitly depend on time t, the total mechanical energy E is conserved:

$$H(q,p) = E$$

Hamilton's principal function $S(q,E,t)$ separates into Hamilton's characteristic function $W(q,E)$ and a time-dependent term:

$$S(q,E,t) = W(q,E) - Et$$

The generalized momentum p is related to the characteristic function W by:

$$p = \frac{dW}{dq}$$

Substituting this expression for momentum into the Hamiltonian yields the time-independent Hamilton-Jacobi equation:

$$\frac{1}{2m}\left(\frac{dW}{dq}\right)^2 + \frac{1}{2}m\omega^2 q^2 = E$$

To solve for $\frac{dW}{dq}$, we subtract the potential energy term from both sides:

$$\frac{1}{2m}\left(\frac{dW}{dq}\right)^2 = E - \frac{1}{2}m\omega^2 q^2$$

Multiplying both sides of the equation by $2m$ gives:

$$\left(\frac{dW}{dq}\right)^2 = 2m\left(E - \frac{1}{2}m\omega^2 q^2\right)$$

Taking the positive square root of both sides yields the differential relation for W:

$$\frac{dW}{dq} = \sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}$$

Integrating both sides with respect to q gives the explicit integral form of Hamilton's characteristic function W:

$$W(q,E) = \int \sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}\, dq$$

Substituting this integral back into the expression for $S(q,E,t)$ gives Hamilton's principal function:

$$S(q,E,t) = \int \sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}\, dq - Et$$

According to Hamilton-Jacobi theory, the transformation to new constant coordinates leads to a new constant momentum $\alpha = E$ and a conjugate constant coordinate β:

$$\beta = \frac{\partial S}{\partial E}$$

Substituting the explicit expression for $S(q,E,t)$ into this relation gives:

$$\beta = \frac{\partial}{\partial E}\left[\int \sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}\, dq\right] - \frac{\partial}{\partial E}(Et)$$

Differentiating the second term with respect to E simplifies the equation to:

$$\beta = \frac{\partial}{\partial E}\left[\int \sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}\, dq\right] - t$$

Interchanging the order of integration and partial differentiation yields:

$$\beta + t = \int \frac{\partial}{\partial E}\left[\sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}\right] dq$$

Evaluating the derivative inside the integral gives:

$$\beta + t = \int \frac{2m}{2\sqrt{2m\left(E - \frac{1}{2}m\omega^2 q^2\right)}}\, dq$$

Cancelling the factor of 2 in the numerator and denominator gives:

$$\beta + t = \int \frac{m}{\sqrt{2mE - m^2\omega^2 q^2}}\, dq$$

Factoring out $m^2\omega^2$ from the square root in the denominator yields:

$$\beta + t = \int \frac{m}{m\omega\sqrt{\frac{2E}{m\omega^2} - q^2}}\, dq$$

Cancelling the mass m from the numerator and denominator simplifies the equation to:

$$\beta + t = \frac{1}{\omega} \int \frac{dq}{\sqrt{\frac{2E}{m\omega^2} - q^2}}$$

Using the standard integration formula $\int \frac{dq}{\sqrt{a^2 - q^2}} = \arcsin\left(\frac{q}{a}\right)$ with $a = \sqrt{\frac{2E}{m\omega^2}}$, the integral evaluates to:

$$\beta + t = \frac{1}{\omega} \arcsin\left(\frac{q}{\sqrt{\frac{2E}{m\omega^2}}}\right)$$

Multiplying both sides by ω gives:

$$\omega(t + \beta) = \arcsin\left(\frac{q}{\sqrt{\frac{2E}{m\omega^2}}}\right)$$

Taking the sine of both sides yields:

$$\sin(\omega t + \omega\beta) = \frac{q}{\sqrt{\frac{2E}{m\omega^2}}}$$

Defining the phase constant as $\phi = \omega\beta$, we have:

$$\sin(\omega t + \phi) = \frac{q}{\sqrt{\frac{2E}{m\omega^2}}}$$

Multiplying both sides by the amplitude term $\sqrt{\frac{2E}{m\omega^2}}$ recovers the complete oscillatory solution for the position $q(t)$:

$$q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \phi)$$

5. Action and Angle Variables

Definition of Action and Angle Variables

For periodic systems with bound motion, action-angle variables provide a powerful technique to determine frequencies of motion without fully integrating the equations of motion. A system undergoes periodic motion if its trajectory in the phase plane (q_i, p_i) forms a closed loop (libration or rotation).

For a conservative system with separable coordinates, the action variable J_i associated with the i-th degree of freedom is defined as the line integral of momentum p_i over one complete period of motion:

$$J_i = \oint p_i \, dq_i$$

Since momentum is defined as $p_i = \frac{\partial W_i(q_i,\alpha)}{\partial q_i}$, the action variable is written as:

$$J_i = \oint \frac{\partial W_i(q_i, \alpha_1, \ldots, \alpha_n)}{dq_i}\, dq_i$$

Evaluating this integral over a full period makes J_i a function solely of the constants of integration $\alpha_1, \alpha_2, \ldots, \alpha_n$:

$$J_i = J_i(\alpha_1, \alpha_2, \ldots, \alpha_n)$$

Inverting these equations allows the constants α_i, and consequently the total energy H, to be expressed entirely in terms of the action variables:

$$H = H(J_1, J_2, \ldots, J_n)$$

We perform a canonical transformation using Hamilton's characteristic function expressed in terms of the new momenta J_i, written as $W(q_1, \ldots, q_n, J_1, \ldots, J_n)$. The coordinate canonically conjugate to J_i is called the angle variable w_i, defined by:

$$w_i = \frac{\partial W(q, J)}{\partial J_i}$$

The equation of motion for the angle variable w_i is given by Hamilton's canonical equation:

$$\dot{w}_i = \frac{\partial H(J)}{\partial J_i}$$

Since the transformed Hamiltonian $H(J)$ depends only on the action variables J_i and not on the angle variables w_i, the time derivative $\dot{w}_i$ is a constant, denoted by ν_i:

$$\nu_i = \frac{\partial H}{\partial J_i}$$

Integrating this equation with respect to time t gives:

$$w_i(t) = \nu_i t + w_i(0)$$

The angle variable w_i increases linearly with time t.

The quantity $\nu_i = \frac{\partial H}{\partial J_i}$ represents the fundamental frequency of the periodic motion in cycles per second (Hertz, Hz). The corresponding angular frequency ω_i in radians per second is given by:

$$\omega_i = 2\pi\nu_i = 2\pi\frac{\partial H}{\partial J_i}$$

Over one complete cycle of coordinate q_i, the change in the angle variable Δw_i is exactly 1:

$$\Delta w_i = \oint \frac{\partial w_i}{\partial q_i}\, dq_i = \oint \frac{\partial^2 W}{\partial q_i \partial J_i}\, dq_i = \frac{d}{dJ_i}\oint p_i\, dq_i = \frac{dJ_i}{dJ_i} = 1$$

Harmonic Oscillator by Action-Angle Variables

Consider again the one-dimensional harmonic oscillator with Hamiltonian:

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2 = E$$

Solving for momentum p in terms of position q and energy E:

$$p = \sqrt{2mE - m^2\omega^2 q^2}$$

The classical turning points occur where momentum $p = 0$:

$$2mE - m^2\omega^2 q^2 = 0$$

Solving for the maximum displacement $q_{\max}$ yields:

$$q_{\max} = \sqrt{\frac{2E}{m\omega^2}}$$

The action variable J is calculated by integrating over one full cycle of oscillation:

$$J = \oint p\, dq = 2\int_{-q_{\max}}^{q_{\max}} \sqrt{2mE - m^2\omega^2 q^2}\, dq$$

Factoring $2mE$ out of the integrand gives:

$$J = 2\sqrt{2mE}\int_{-q_{\max}}^{q_{\max}} \sqrt{1 - \frac{m\omega^2 q^2}{2E}}\, dq$$

We substitute $q = \sqrt{\frac{2E}{m\omega^2}}\sin\theta$, which gives differential $dq = \sqrt{\frac{2E}{m\omega^2}}\cos\theta\, d\theta$. The limits of integration change from $q = -q_{\max}, q_{\max}$ to $\theta = -\frac{\pi}{2}, \frac{\pi}{2}$:

$$J = 2\sqrt{2mE}\sqrt{\frac{2E}{m\omega^2}}\int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta$$

Simplifying the factor outside the integral:

$$J = \frac{4E}{\omega}\int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta$$

Using the trigonometric identity $\cos^2\theta = \frac{1+\cos(2\theta)}{2}$:

$$J = \frac{4E}{\omega}\int_{-\pi/2}^{\pi/2} \frac{1+\cos(2\theta)}{2}\, d\theta$$

Integrating term by term yields:

$$J = \frac{2E}{\omega}\left[\theta + \frac{\sin(2\theta)}{2}\right]_{-\pi/2}^{\pi/2}$$

Evaluating at the limits gives:

$$J = \frac{2E}{\omega}\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right)$$

$$J = \frac{2\pi E}{\omega}$$

Solving this relation for the total energy $E = H$:

$$H(J) = \frac{\omega J}{2\pi}$$

Expressed in terms of frequency in cycles per second, $\nu = \frac{\omega}{2\pi}$:

$$H(J) = \nu J$$

The frequency of the motion is calculated using the derivative of H with respect to J:

$$\frac{\partial H}{\partial J} = \frac{\partial(\nu J)}{\partial J} = \nu$$

The corresponding angular frequency ω is:

$$\omega = 2\pi\frac{\partial H}{\partial J} = 2\pi\left(\frac{\omega}{2\pi}\right) = \omega$$

The time evolution of the angle variable w is determined by Hamilton's equation:

$$\dot{w} = \frac{\partial H}{\partial J} = \nu$$

Integrating with respect to time t:

$$w(t) = \nu t + w(0)$$

This result demonstrates that the angle variable $w(t)$ advances linearly in time.

6. Kepler Problem in Action-Angle Variables

Consider a particle of reduced mass μ moving in a central gravitational potential $V(r) = -\frac{k}{r}$, where $k = GM\mu$. In spherical polar coordinates (r, θ, ϕ), the Lagrangian kinetic energy is

$$T = \frac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2\right)$$

The canonical momenta are given by

$$p_r = \mu \dot{r}$$

$$p_\theta = \mu r^2 \dot{\theta}$$

$$p_\phi = \mu r^2 \sin^2\theta \dot{\phi}$$

The Hamiltonian of the system expressed in terms of these momenta is

$$H = \frac{1}{2\mu}\left(p_r^2 + \frac{p_\theta^2}{r^2} + \frac{p_\phi^2}{r^2 \sin^2\theta}\right) - \frac{k}{r} = E$$

Hamilton's characteristic function $W(r, \theta, \phi)$ separates as

$$W(r, \theta, \phi) = W_r(r) + W_\theta(\theta) + W_\phi(\phi)$$

Since ϕ is a cyclic coordinate, the momentum p_ϕ is a constant of motion denoted by α_ϕ:

$$p_\phi = \frac{dW_\phi}{d\phi} = \alpha_\phi$$

The action variable J_ϕ is defined as

$$J_\phi = \oint p_\phi d\phi$$

Substituting $p_\phi = \alpha_\phi$ into the integral over a complete revolution from 0 to 2π gives

$$J_\phi = \int_0^{2\pi} \alpha_\phi d\phi$$

$$J_\phi = 2\pi\alpha_\phi$$

Next, consider the θ-dependent part of the Hamilton-Jacobi equation:

$$p_\theta^2 + \frac{\alpha_\phi^2}{\sin^2\theta} = \alpha_\theta^2$$

where α_θ is the total orbital angular momentum.

Solving for p_θ gives

$$p_\theta = \frac{dW_\theta}{d\theta} = \sqrt{\alpha_\theta^2 - \frac{\alpha_\phi^2}{\sin^2\theta}}$$

The action variable J_θ is defined as

$$J_\theta = \oint p_\theta d\theta$$

Substituting p_θ into the integral yields

$$J_\theta = \oint \sqrt{\alpha_\theta^2 - \frac{\alpha_\phi^2}{\sin^2\theta}}\, d\theta$$

Evaluating this integral over a complete cycle of libration in θ gives

$$J_\theta = 2\pi(\alpha_\theta - \alpha_\phi)$$

Adding $J_\phi = 2\pi\alpha_\phi$ to J_θ yields

$$J_\theta + J_\phi = 2\pi\alpha_\theta$$

Now, consider the radial part of the Hamilton-Jacobi equation:

$$\frac{1}{2\mu}\left[\left(\frac{dW_r}{dr}\right)^2 + \frac{\alpha_\theta^2}{r^2}\right] - \frac{k}{r} = E$$

Solving for radial momentum $p_r = \frac{dW_r}{dr}$ gives

$$p_r = \sqrt{2\mu E + \frac{2\mu k}{r} - \frac{\alpha_\theta^2}{r^2}}$$

The radial action variable J_r is defined as

$$J_r = \oint p_r dr$$

Substituting p_r into the phase integral yields

$$J_r = \oint \sqrt{2\mu E + \frac{2\mu k}{r} - \frac{\alpha_\theta^2}{r^2}}\, dr$$

For bound motion ($E < 0$), evaluating this integral using complex contour integration gives

$$J_r = 2\pi\left(\frac{\mu k}{\sqrt{-2\mu E}} - \alpha_\theta\right)$$

Substituting $2\pi\alpha_\theta = J_\theta + J_\phi$ into the expression for J_r gives

$$J_r = \frac{2\pi\mu k}{\sqrt{-2\mu E}} - (J_\theta + J_\phi)$$

Rearranging terms to isolate the energy dependent term on one side yields

$$J_r + J_\theta + J_\phi = \frac{2\pi\mu k}{\sqrt{-2\mu E}}$$

Squaring both sides of the equation gives

$$(J_r + J_\theta + J_\phi)^2 = \frac{4\pi^2\mu^2k^2}{-2\mu E}$$

Simplifying the fraction on the right-hand side yields

$$(J_r + J_\theta + J_\phi)^2 = \frac{2\pi^2\mu k^2}{-E}$$

Solving for energy $E = H$ gives the Hamiltonian entirely in terms of the sum of action variables:

$$H = -\frac{2\pi^2\mu k^2}{(J_r + J_\theta + J_\phi)^2}$$

The fundamental frequencies ν_i are obtained by taking partial derivatives of H with respect to each action variable J_i:

$$\nu_r = \frac{\partial H}{\partial J_r} = \frac{4\pi^2\mu k^2}{(J_r + J_\theta + J_\phi)^3}$$

$$\nu_\theta = \frac{\partial H}{\partial J_\theta} = \frac{4\pi^2\mu k^2}{(J_r + J_\theta + J_\phi)^3}$$

$$\nu_\phi = \frac{\partial H}{\partial J_\phi} = \frac{4\pi^2\mu k^2}{(J_r + J_\theta + J_\phi)^3}$$

Comparing these expressions shows that all three frequencies are identical:

$$\nu_r = \nu_\theta = \nu_\phi$$

Since the three fundamental frequencies are equal, the system exhibits complete spatial degeneracy (triply degenerate). Because the radial frequency equals the angular frequencies, the time taken for the particle to return to its radial turning point is identical to the time taken to complete one full revolution in angle, which ensures that the Kepler orbit is closed.

Questions

Short form first (2–3 marks), then long form (5–10 marks). **No Assam University questions were found for this unit in the paper bank** — the questions below are from other universities' papers and standard syllabus-derived questions, marked as such.

Short-form questions

S1. [standard] State the Hamilton-Jacobi equation.

Answer in brief: Section 1 — $H(q, \partial S/\partial q, t) + \partial S/\partial t = 0$, where S is Hamilton's principal function.

S2. [standard] Define Hamilton's characteristic function W.

Answer in brief: Section 2 — for a time-independent Hamiltonian, $S = W(q, \alpha) - Et$ and W satisfies $H(q, \partial W/\partial q) = E$.

S3. [standard] Define action and angle variables.

Answer in brief: Section 5 — $J = \oint p\, dq$ over one period; $w = \partial W/\partial J$ increases uniformly with time; $H = H(J)$ only.

S4. [standard] What is the frequency of a harmonic oscillator in terms of its action variable?

Answer in brief: Section 5 — $J = \oint p\, dq = 2\pi E/\omega$, so $H = \nu J$ with $\nu = \omega/2\pi$ cycles per second; the angular frequency is $\omega = 2\pi\, \partial H/\partial J$.

Long-form questions

L1. [non-AU] Solve the linear harmonic oscillator using the Hamilton-Jacobi formulation. — Christ College, M.Sc Sem-I, CC19PPHY1C01 Classical Mechanics, Nov 2025, Section B Q11.

What to write: Section 4 in full — the Hamilton-Jacobi equation for $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, the characteristic function W, and $q(t)$ from $\beta = \partial S/\partial E$, giving $q = \sqrt{2E/m\omega^2}\,\sin(\omega t + \phi)$.

L2. [non-AU] Calculate the frequency of a linear harmonic oscillator by using the method of action-angle variables. — Christ College Nov 2025, Section C Q17.

What to write: Section 5 in full — $J = \oint p\, dq = 2\pi E/\omega$, so $H = \nu J$ and the frequency follows from $\nu = \partial H/\partial J = \omega/2\pi$ cycles per second (angular frequency ω).

L3. [non-AU] Apply Hamilton-Jacobi theory to determine the equation of motion of a body falling vertically in a uniform gravitational field. — St. Aloysius, M.Sc Sem-I, Nov 2021, Part B Q17.

What to write: Sections 1–2 — with $H = p^2/2m + mgy$, the Hamilton-Jacobi equation gives S, and $\beta = \partial S/\partial E$ yields $y(t)$ for uniformly accelerated fall, $\ddot{y} = -g$.

L4. [standard] Solve the Kepler problem in action-angle variables: obtain the action variables, the Hamiltonian in terms of them, and the frequencies of motion. Also state the Hamilton-Jacobi equation, define Hamilton's characteristic function, and explain separation of variables.

What to write: Sections 1–3 and 6 in full — the Hamilton-Jacobi equation and W, additive separation, the three action variables, $H = -2\pi^2\mu k^2/(J_r + J_\theta + J_\phi)^2$, and the three equal frequencies $\nu_i = \partial H/\partial J_i$ (degeneracy, closed orbit).

Sources

These notes were written from the unit syllabus only, as the user asked. No external PDF, textbook, or lecture notes were used. The questions are from the Assam University paper bank and other universities' papers (see the Questions section above), not from any source used for the notes.